Chapter 5. Ratio Estimation

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1 Auxiliary Information

So far the data were modeled as $y_i = \mu + \varepsilon_i$ with $\mathrm{E}(\varepsilon_i) = 0$, so $\mathrm{E}(y_i) = \mu$ for every unit. If an auxiliary variable x related to y is available for each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and knowledge of the population mean μ_x or total τ_x of x can be used to sharpen the estimate of μ_y or τ_y. This chapter uses a line through the origin (ratio estimation); Ch. 7 uses a general line (regression estimation).

Example 5.1: Sugar in a truckload of oranges

The wholesale price of oranges is based on the total sugar content τ_y of the load. The estimate $N\bar{y}$ needs the number of oranges N, which is unknown. But sugar content y is closely related to weight x, and the total weight τ_x of the truckload is easy to obtain. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\tau_x,$$

which does not require N.

Other ratios of interest: total monthly food budget to total monthly income per family; number of children to number of residents; average amount of a new drug needed to evoke a response relative to that of a standard drug.

2 Ratio Estimation under SRS

Motivation. If an auxiliary variable x with known population total is strongly correlated with y, the ratio $\bar{y}/\bar{x}$ is more stable than $\bar{y}$ itself. Multiplying it by the known total of x gives a ratio estimator with much smaller variance.

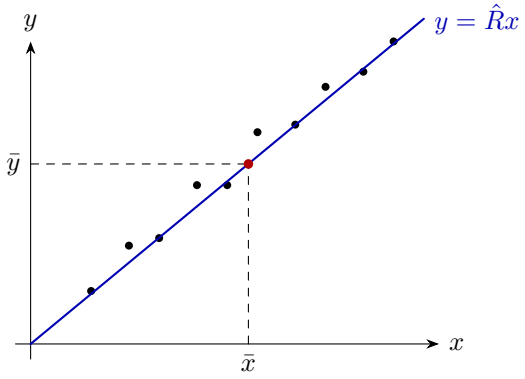


Figure. *Ratio estimation fits a line through the origin with slope $\hat{R} = \bar{y}/\bar{x}$. When y is roughly proportional to x, $\hat{\tau}_y = \hat{R}\,\tau_x$ uses the known total τ_x and has much smaller variance than $N\bar{y}$.*

Let (x_i, y_i), $i = 1, \dots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - r x_i)^2$,

ratio:	$r = \bar{y}/\bar{x}$,	$\widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} \cdot \dfrac{1}{\mu_x^2}\, s_r^2$	(μ_x replaced by $\bar{x}$ if unknown);
total:	$\hat{\tau}_y = r\tau_x$,	$\widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \dfrac{1-f}{n} \dfrac{s_r^2}{\bar{x}^2}$;	
mean:	$\hat{\mu}_y = r\mu_x$,	$\widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \dfrac{1-f}{n} s_r^2$.	

The residuals $y_i - r x_i$ measure the departure from a line through the origin. If $y_i \approx r x_i$ for all units, s_r^2 is small and the ratio estimator is very precise.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^{N}(y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x} \cdot \frac{1}{n}\sum_{i=1}^{n}(y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^{N}(y_i - R x_i)^2 = \sum_{i=1}^{N}\left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum(x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = Rx_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n} \cdot \frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is [1.0511, 1.0809].

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

Motivation. A ratio of unbiased estimators is not unbiased. The bias is of order $1/n$, smaller than the standard error, so in moderate samples it is negligible.

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1+\delta_y)(1+\delta_x)^{-1} = R(1+\delta_y)(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):

$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x}-\mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x}-\mu_x)(\bar{y}-\mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1-f}{n}S_x^2 - \frac{R}{\mu_x \mu_y}\frac{1-f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ■

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho\frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum(x_i-\mu_x)\,\beta(x_i-\mu_x)}{\frac{1}{N-1}\sum(x_i-\mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r,\bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r)-R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r,\bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r,\bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r)-R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the standard error whenever the coefficient of variation of $\bar{x}$ is small, which holds for moderately large n.

4 Ratio Estimation in Stratified Random Sampling

Motivation. Ratio estimation and stratification can be combined, either one ratio per stratum (separate) or one ratio for the whole sample (combined). The choice depends on stratum sample sizes and on whether the ratios differ across strata.

With strata $h = 1, \ldots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h\, r_h\, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C\,\mu_x, \qquad r_C = \frac{\bar{y}_{\text{st}}}{\bar{x}_{\text{st}}}, \quad \bar{y}_{\text{st}} = \sum_h w_h \bar{y}_h,\ \bar{x}_{\text{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\text{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h}(y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\text{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\text{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\text{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$\begin{aligned}(N_h - 1)S_{h,S}^2 &= \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h - 1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h\rho_h S_{xh} S_{yh}),\\ (N_h - 1)S_{h,C}^2 &= \sum_i \left[(y_{hi} - \mu_{yh}) - R_C(x_{hi} - \mu_{xh})\right]^2\\ &= (N_h - 1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C\rho_h S_{xh} S_{yh}).\end{aligned}$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\text{st}} - R_C\bar{x}_{\text{st}} = \bar{d}_{\text{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C\mu_{xh}$.) Subtracting,

$$\begin{aligned}S_{h,C}^2 - S_{h,S}^2 &= (R_C^2 - R_h^2)S_{xh}^2 - 2(R_C - R_h)\rho_h S_{xh} S_{yh}\\ &= (R_C - R_h)^2 S_{xh}^2 + 2(R_C - R_h)(R_h S_{xh}^2 - \rho_h S_{xh} S_{yh}).\end{aligned}$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\text{Var}(\hat{\mu}_{yRC}) - \text{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1 - f_h}{n_h}(R_C - R_h)^2 S_{xh}^2 \ \geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But each r_h carries its own ratio bias, and these add up across strata. When the n_h are small the separate estimator can be badly biased; when the n_h are large (over about 50) the bias is negligible. With small strata the combined estimator is safer.

5 Ratio Estimator versus the Sample Mean

Motivation. The ratio estimator beats $\bar{y}$ when the correlation between x and y exceeds half the ratio of their coefficients of variation; this condition tells us when the auxiliary information is worth using.

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{Var}(\hat{\theta}_1)/\mathrm{Var}(\hat{\theta}_2)$:

$$\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2, \qquad \mathrm{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\mathrm{RE}(\bar{y}, \hat{\mu}_y) = \frac{\mathrm{Var}(\bar{y})}{\mathrm{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\mathrm{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2} \cdot \frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2} \cdot \frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\mathrm{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ■

The ratio estimator is more efficient than $\bar{y}$ when x and y are highly positively correlated; the estimated relative efficiency is meaningful only for large samples.

6 Selected exercises

Motivation. The ratio estimator was justified above by design-based approximations. The first item shows that it is also the exact optimal estimator under a natural model, and the second is a complete computation for a total.

Exercise 5.1 (The ratio estimator as a weighted least-squares fit)

Consider the model $y_i = \beta x_i + \varepsilon_i$, $i = 1, \dots, n$, with fixed $x_i > 0$, $\mathrm{E}(\varepsilon_i) = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i$ and uncorrelated errors.
(a) Show that $r = \sum y_i / \sum x_i$ minimizes the weighted sum of squares $\sum_i (y_i - \beta x_i)^2 / x_i$.
(b) Show that r is unbiased for β with $\mathrm{Var}(r) = \sigma^2 / \sum_i x_i$, and that it has the smallest variance among all linear unbiased estimators $\sum_i a_i y_i$ of β.

Solution. [Supplement]

(a) $Q(\beta) = \sum_i (y_i - \beta x_i)^2 / x_i$ is a quadratic in β with $Q'(\beta) = -2\sum_i (y_i - \beta x_i) = -2(\sum_i y_i - \beta \sum_i x_i)$ and $Q'' = 2\sum_i x_i > 0$, so the unique minimizer is $\beta = \sum y_i / \sum x_i = r$.

(b)

Step 1. (Unbiasedness and variance.) $r = \sum_i a_i^0 y_i$ with $a_i^0 = 1/\sum_j x_j$. Then $\mathrm{E}(r) = \beta \sum_i a_i^0 x_i = \beta$ and $\mathrm{Var}(r) = \sigma^2 \sum_i (a_i^0)^2 x_i = \sigma^2 / \sum_i x_i$.

Step 2. (Unbiasedness constraint.) A linear estimator $\sum_i a_i y_i$ has mean $\beta \sum_i a_i x_i$, so it is unbiased for every β iff $\sum_i a_i x_i = 1$. Its variance is $\sigma^2 \sum_i a_i^2 x_i$.

Step 3. (Cauchy–Schwarz.) Writing $a_i x_i = (a_i\sqrt{x_i})\sqrt{x_i}$,

$$1 = \Big(\sum_i a_i x_i\Big)^2 \le \Big(\sum_i a_i^2 x_i\Big)\Big(\sum_i x_i\Big),$$

so $\sigma^2 \sum_i a_i^2 x_i \ge \sigma^2 / \sum_i x_i = \mathrm{Var}(r)$, with equality iff $a_i\sqrt{x_i} \propto \sqrt{x_i}$, i.e. a_i constant, i.e. $a_i = a_i^0$.

Thus the ratio estimator is the best linear unbiased estimator when y is proportional to x with error variance proportional to x, the situation of the oranges example. ■

Exercise 5.2 (Sales of television sets)

A corporation wants the total earnings τ_y from sales of television sets in the current three-month period. An SRS of $n = 13$ of its $N = 123$ district offices gives the previous-year figures x_i and current figures y_i below, and the previous-year total for all offices is $\tau_x = 128{,}200$.

x_i	550	720	1500	1020	620	980	928	1200	1350	1750	670	729	1530
y_i	610	780	1600	1030	600	1050	977	1440	1570	2210	980	865	1710

Estimate τ_y by the ratio estimator and place a bound on the error.

Solution.

Step 1. $\sum x_i = 13{,}547$, $\sum y_i = 15{,}422$, so $r = 15{,}422/13{,}547 = 1.1384$ and $\hat\tau_y = r\tau_x = 1.1384 \times 128{,}200 = 145{,}944$.

Step 2. The residual variance is $s_r^2 = \frac{1}{12}\sum_i (y_i - r x_i)^2 = 12{,}990$, and $\mu_x = \tau_x/N = 1042.3$ is known.

Step 3. By the ratio-estimator theorem,

$$\widehat{\mathrm{Var}}(\hat\tau_y) = \tau_x^2 \frac{1-f}{n}\frac{s_r^2}{\mu_x^2} = N^2 \frac{1-f}{n} s_r^2 = 123^2 \cdot \frac{1 - 13/123}{13} \cdot 12{,}990 = 1.352 \times 10^7,$$

so the bound is $2\sqrt{1.352 \times 10^7} = 7354$ and τ_y is estimated by $145{,}944 \pm 7354$.

Step 4. For comparison, $N\bar y = 123 \times 1186.3 = 145{,}916$ has $s_y^2 = 231{,}543$ and bound

$$2N\sqrt{(1-f)s_y^2/n} \approx 31{,}050:$$

the ratio estimator is about four times as precise, since current and previous sales are almost proportional. ■